Rotary Flow Sensing & Power Generation

Watts Water Technologies

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1. Overview & Background

This document is a master reference for **rotary flow sensing and micro-hydro power generation in the 0 to 4 GPM range**, covering technology selection, bench characterization, and system architectures for fixtures that need flow data, self-generated power, or both.

Two paths are covered:

- **Flow sensing alone**, using a dedicated rotary flow meter where wired power or a replaceable battery is acceptable.
- **Flow sensing + power generation**, using a micro-hydro turbine generator where the fixture must be sealed, long-life, or self-powered (most commonly in IoT-enabled products). The same turbine can optionally double as the flow sensor.

Most of the document focuses on the second path: classifying turbine generators, characterizing them on the bench, and matching them to product requirements.

Design priorities across all applications:

- Low cost
- Accurate at low flow rates
- Low power consumption
- Potential for self-powered operation (no wired power, no battery replacement)

1.1 Why Rotary? Why 0 to 4 GPM?

The scope is **0 to 4 GPM**, covering bottle fillers, drinking stations, kitchen sinks, restroom sinks, residential showers and other low-flow applications. Single fixtures are typically below 2 GPM; the 4 GPM ceiling accommodates multi-fixture configurations (e.g., up to four sinks on one sensor). Within this band, **0 to 1 GPM is treated as ultra-low flow**, where startup torque, bearing drag, and electromagnetic braking dominate behavior. Different turbines are best at different sub-ranges (some peak around 0.25 to 0.33 GPM, others closer to 1 GPM or higher), and matching that sub-range to the application is a central job of the selection framework.

Rotary sensing measures volume directly by counting spinner revolutions rather than inferring flow from pressure. At low flow, ΔP across any restriction is too small to measure reliably; a spinning turbine at 0.5 GPM produces a clean, countable pulse train. Above 4 GPM, differential pressure sensing starts taking over.

1.2 Prior Work

An initial IoT Technology Readiness Report at Bradley benchmarked and tore down competitor faucets (U by Moen, Delta Essa VoiceIQ), independently confirming the **GEMS 238600** turbine in both and validating it as the optimal sensing element. That report identified differential pressure sensing as viable for higher-flow products (8 GPM +) and recommended continued research into turbine generators for self-powered applications, which is where this document picks up.

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2. Flow Sensing

2.1 Technology Survey

Technology	Principle	Strengths	Weaknesses	Best Fit
Rotary (Turbine / Paddle)	Spinning element; RPM $\propto$ flow	Low cost, good low-flow accuracy	Moving parts, wear	0.08 to 8 GPM

Technology	Principle	Strengths	Weaknesses	Best Fit
Positive Displacement	Fluid fills/empties fixed chambers	Good at low flow	Many moving parts, high ΔP	0.25 to 2 GPM
Differential Pressure	ΔP across restriction ∝ Q^2	No moving parts, cheap	Poor at low flow	>4 GPM
Ultrasonic	Transit time of sound	Non-invasive, no ΔP	Expensive, special housing	High accuracy
Magnetic	Faraday's law on conductive fluid	Non-invasive, accurate	Needs conductive fluid	Industrial
Thermal / Coriolis	Heat transfer or mass flow	Very accurate	Expensive, power-hungry	Out of scope
Vortex Shedding	Vortices behind bluff body	Accurate, no moving parts	Min flow >2 GPM, high ΔP	>2 GPM
Vane / Piston	Spring-loaded vane deflects with flow	Wide range, low-flow capable	Mechanical complexity	Similar to turbine

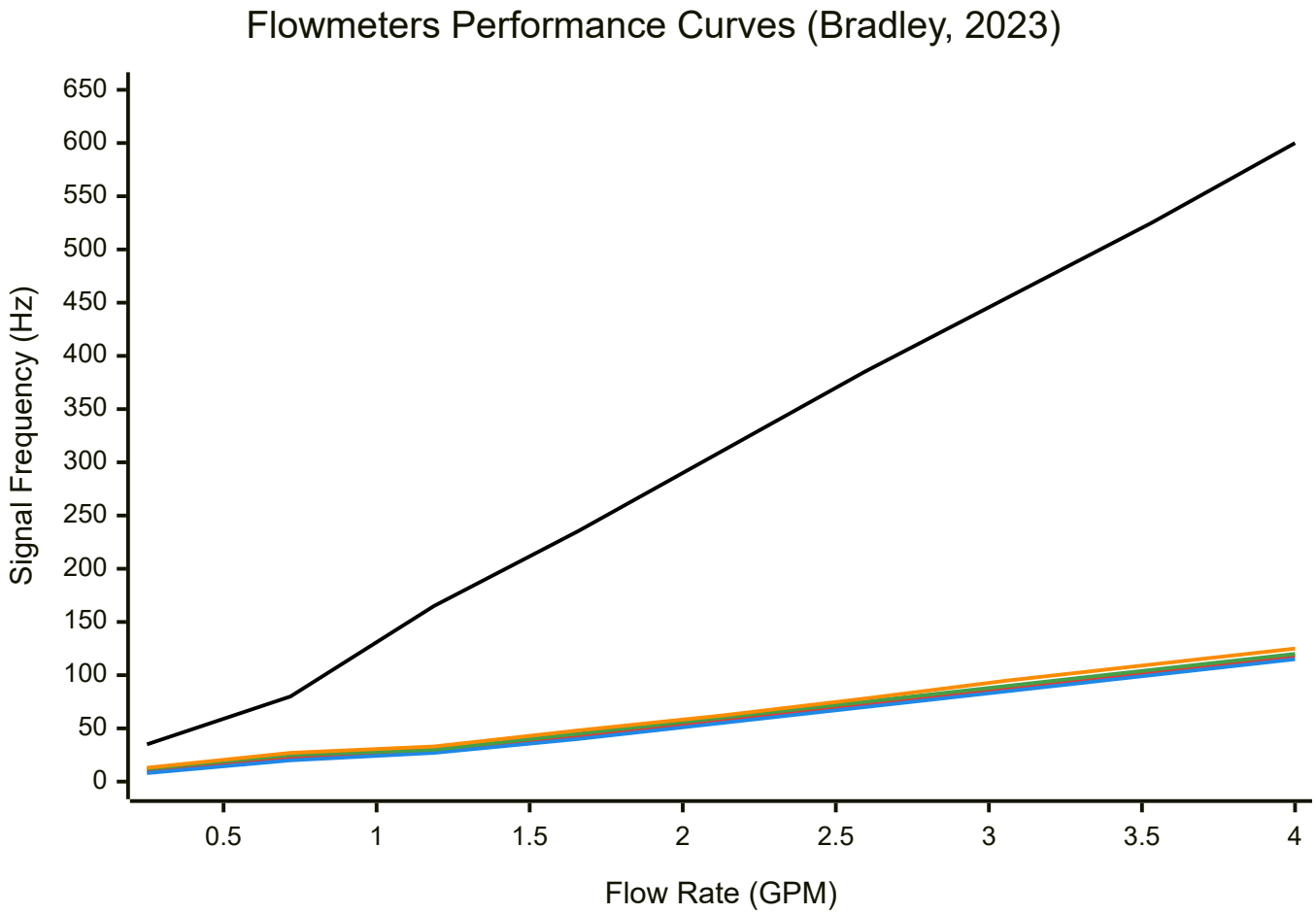
2.2 Rotary Sensors: Radial vs. Axial

Type	Description	Min Flow	Notes
Radial / Paddle Wheel	Only part of rotor in flow path	~0.1 to 0.25 GPM	Cheaper, less accurate at low flow
Axial / Inline	Full propeller inline; Hall reads blades	~0.05 to 0.08 GPM	Better low-flow sensitivity

Paddle-wheel meters (Gredia, Sea YF-S201, Uxcell) require higher flow to start spinning. Axial turbines (GEMS 238600, Sika VY10, Auk Mueller MT5) offer superior low-flow performance. The **GEMS 238600** was selected as the primary sensing element based on in-house testing and competitor validation.

Flowmeter Performance Curves

Signal frequency vs. flow rate is linear ($f = k \cdot Q$) for all rotary meters. The GEMS 238600 produces roughly **4 to 5x the frequency-per-GPM** of any other meter tested, which is the primary reason it was selected: more pulses per unit volume means higher resolution at low flow.



Legend:

	Meter	Slope (Hz / GPM)
	GEMS 238600	~150
	Sea YF-S201	~30
	Gredia FS200A	~29
	Sika VY10	~31
	Auk Mueller MT5	~32

Data reproduced from Bradley in-house testing, 2023.

2.3 Hall Effect Sensors

The turbine's embedded magnet is read by an external Hall effect sensor. The GEMS 238600 emits approximately **100 Gauss** at the sensor zone (measured in-house).

Sensor	Supply Voltage	Active Current	Active Power	Sleep Current	B_{OP}	B_{RP}	Key Advantage
Allegro A1220	3 to 24 V	4 mA (always on)	~12 mW @ 3 V	N/A	35 G typ	15 G typ	Proven, wide voltage range
Allegro A1171	1.65 to 3.5 V	3.5 to 12 µA	~15 to 42 µW	8 µA	45 G typ	30 G typ	~800x lower power than A1220

Both operate comfortably with the 100 G field from the GEMS turbine. The **A1171 is preferred for any battery-powered or energy-harvesting application**.

> *Aside:* the A1171 has not yet been bench-validated against a live GEMS turbine in our setup. The 100 G field comfortably exceeds its typical B_{OP} of 45 G on paper, but a confirmation pass with the live turbine should happen before designing it into a product.

> **Shortcut for simple cases:** if the product allows wired power or a replaceable battery, the answer is GEMS 238600 + A1220 (wired) or A1171 (battery). Everything that follows is for the harder case: sealed, long-life, or self-powered deployments where a turbine generator is needed.

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3. Turbine Generators, Concepts

This section covers what generators exist and how they work. Bench testing and selection rules come in §4.

3.1 Goals & Operating Range

The goal is to generate usable electrical power from water flow anywhere in the 0 to 4 GPM band, with particular attention to ultra-low flow (0 to 1 GPM) where most off-the-shelf generators struggle. Generators can also double as flow sensors since output frequency and power both scale with flow rate (covered in §6).

3.2 Magnet Magnetization

All rotor magnets must be **diametrically magnetized** (N/S poles on opposite sides, perpendicular to the spin axis). As the magnet spins, the field alternates and induces voltage in the stator coils. An axially magnetized magnet spinning on its own axis produces no flux change and no voltage. This is not a design variable; all generators in this catalogue use diametric magnetization.

For low-flow turbines the stator usually surrounds the rotor; partial-admission designs sometimes invert this with the rotor outside the stator.

3.3 Admission Types

Both full and partial admission use a directed jet, but differ in geometry and performance:

Type	Description	Key Characteristic
Full Admission	Tight-tolerance nozzle directs jet around entire rotor circumference; all blades in flow	Lower startup flow, higher ΔP, precise machining
Partial Admission	Simpler, smaller inlet hole; jet hits only part of rotor at a time	Higher startup flow, lower ΔP, simpler geometry

Full admission generators can start generating at lower flow rates (critical for ultra-low-flow targets) because of a stronger jet and/or lighter rotor.

3.4 Coil Configurations & Flux Types

Flux Types

	Axial Flux	Radial Flux
Flux vs rotation axis	Parallel (along spin axis)	Perpendicular (across spin axis)
Typical coil shape	Pancake (claw pole) or wrapped cylinder (belt)	Small rings pointing inward, or ring-in-ring
Flux guides needed?	Yes, always	Typically no
Cycles per revolution	Magnet pole pairs × claw-pole pairs	Magnet pole pairs

Coil Configurations

Our Name	Literature Name	Flux	Description	Example
Claw pole	Claw pole	Axial	Pancake coil above soft-iron claws that funnel flux through the coil	Zurn P6900
Belt	Axial flux wrapped	Axial	Coil wraps around rotor cylinder, enclosing the	Toto EcoPower, Toto 10s Dynamo

Our Name	Literature Name	Flux	Description	Example
			magnet. Best flux coupling of the axial types	(EDV462/EDV561)
Spoke	Hub generator	Radial	Multiple small coils facing inward; magnet ring spins outside (outrunner)	F50
Gyro	Radial flux ring	Radial	Ring inside another ring; streamlined for in-pipe mounting	M6 axial propeller

Belt Sub-Categories

- **Single coil:** one electrical circuit, two wires (e.g., Toto EcoPower).
- **Dual coil:** two independent circuits, four wires. One coil dedicated to always-on critical functions (solenoid, MCU); the other powers shed-able functions (display, radio). The Toto 10s Dynamo uses this with a high-inductance secondary coil. Toto advertises that 10 seconds of flow fully initializes the electronics and sustains power briefly after flow stops for clean valve closure. Tradeoff: high inductance increases rotor braking.

Three-Phase (Spoke / Hub)

Spoke designs use 3n non-overlapping coils like a motor. Six diodes confirm 3-phase generation (3-phase bridge rectifier). Smoother power delivery and even braking, but reduced ripple is a disadvantage if ripple frequency is being used to infer RPM (see §6.3).

3.5 Physics: Power Generation

Induced EMF

The peak induced EMF in a rotating coil is

$$\varepsilon_{peak} = NBA\omega$$

where N is the number of coil turns, B is the magnet field strength, A is the coil area, and ω is the rotor angular velocity. The same NBA product also sets the magnetic braking torque on the rotor, which is why every variable that increases output voltage also increases electromagnetic braking. Two levers help without that penalty:

- **Thicker wire (lower R_{coil}):** less power dissipated as heat internally, more delivered to the load.
- **More water pressure:** more torque to overcome braking. Pure energy input.

What Drives Output Power

In the **linear region** (moderate ω, no magnetic saturation, rotor not yet braking-limited) output power scales with V^2 and therefore approximately with ω^2 and Q^2. At higher flow rates this breaks down: voltage flattens as electromagnetic braking, bearing drag, and hydrodynamic losses outpace available torque, so power rises sub-quadratically and asymptotes.

Why Each Generator Has a Sweet Spot

At very low flow, friction and bearing losses consume most of the available hydraulic power. As flow increases, output rises. Electromagnetic braking also rises with rotor speed, and at high flow the braking torque becomes significant enough that additional water energy is increasingly wasted. **For most turbines, efficiency therefore peaks at an intermediate flow rate** (the turbine's natural sweet spot). Some turbines, notably axial propellers like the M6, show no interior peak and are simply most efficient at the lowest flow they can support. §4 turns this distinction into the R_{eff} / R_{max} selection rule.

Pressure vs. Flow Rate

$$P_{hydraulic} = \Delta P \cdot Q$$

ΔP provides torque; Q provides rotor speed. Both are needed. Output power rises with both at a fixed load but is bounded by the rotor's mechanical limits.

Efficiency

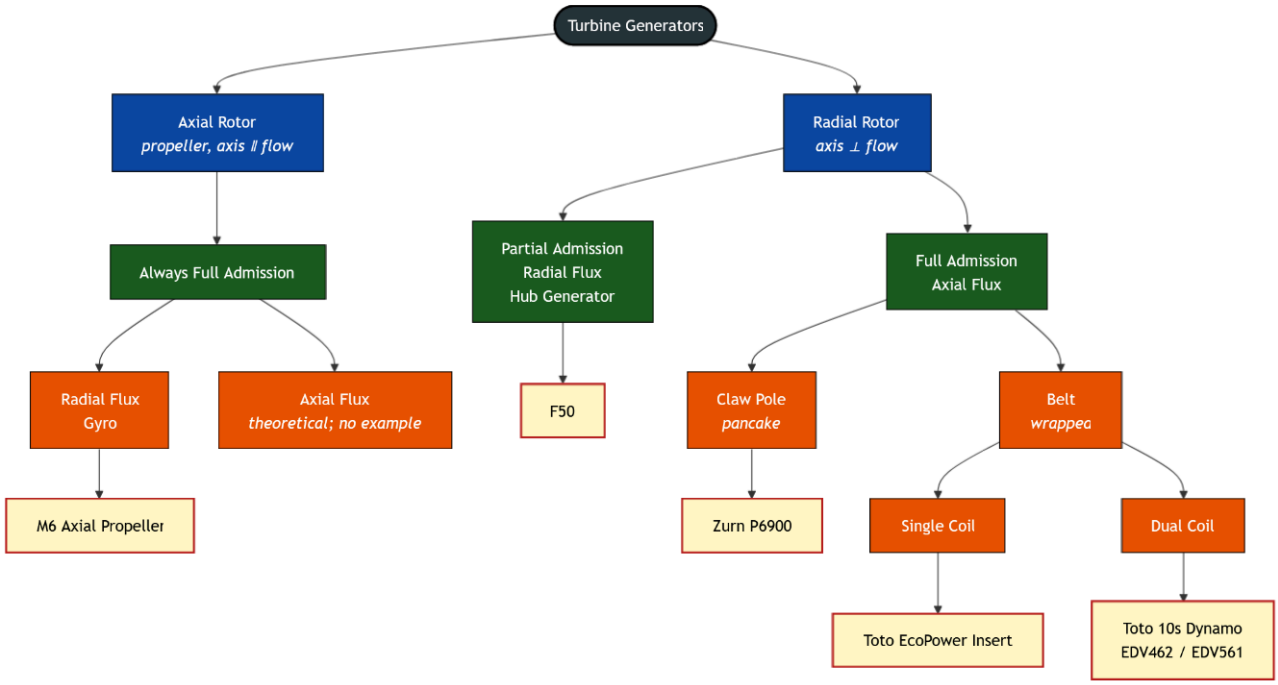
$$\eta = \frac{P_{electrical}}{P_{hydraulic}} = \frac{P_{load}}{\Delta P \cdot Q}$$

Since neither term is linear with flow, the efficiency sweet spot is not predictable from geometry alone, hence the bench testing in §4.

3.6 Classification System & Family Tree

Turbines are classified by **rotor orientation × admission × flux direction × coil configuration**. These axes are largely independent.

- **Axial rotor (propeller):** spin axis parallel to flow. Always full admission.
- **Radial rotor:** spin axis perpendicular to flow. Can be full or partial admission.



Legend: rotor type · admission · flux / coil configuration · commercial example

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4. Bench Testing & Load Selection

The goal of testing is twofold: (1) characterize each turbine's general flow behavior (usable flow range, Q_{start}, ΔP vs. Q, and where the efficiency peak sits if there is one), and (2) pick **one fixed load resistor per turbine** so the rest of the design can treat the turbine as a fixed component. A pre-programmed dynamic load (a small lookup table of resistors switched by flow rate) is a viable alternative that adds a few dollars in BOM and is worth considering for turbines whose optimal R shifts a lot across the operating range; see §4.2 for the comparison and §5.1 Step 2 for a worked example showing how much benefit it actually buys.

> Detailed test rig, procedures, and per-turbine raw data are maintained in a separate **Flow Sensing & Power Generation, Testing & Characterization** document.

4.1 What Gets Measured

Per turbine, once:

- R_{coil}, DC coil resistance
- Teardown geometry: coil ID/OD, pole/claw count, wire gauge. Number of turns N is **estimated** from coil cross-section, wire gauge, and a packing-factor assumption rather than measured directly (counting turns destructively is impractical and not very accurate).

Per flow rate Q, swept 0 to 4 GPM:

- ΔP under load
- Q_{start}, the minimum flow producing any open-circuit voltage
- ΔP_{oc} at open circuit (optional, see note below)
- Full R_{load} sweep, recording V_{load}, current, AC frequency f, and computed power and efficiency

> **On ΔP_{oc}:** open-circuit ΔP vs. Q characterizes the purely hydraulic loss with no electromagnetic braking. Because measured efficiencies are typically under 10% (often under 5%), the electromagnetic component is a small fraction of the total ΔP, and ΔP_{oc} is usually within a few percent of ΔP under load. It's a sanity check, not a critical measurement, and can be skipped on turbines where bench time is limited.

> B (magnet field) and L (inductance) are not directly measurable with current equipment; flux coupling is treated as a single lumped experimental constant from the open-circuit sweep.

> Regulated-DC units (e.g., Zurn P6900) are tested by tapping the raw coil leads upstream of the internal regulator.

4.2 The R_{eff} / R_{max} Rule

Picking the resistance that produces the most raw power at a single flow rate is tempting but wrong: that resistance is only optimal at that one flow rate, and at other flow rates within the operating range it gives up significant output. The useful choice is the resistance that balances power output against hydraulic cost across the **whole expected range of flow**, which corresponds to the efficiency peak.

Symbol	Name	Definition
R_{eff}	Peak-efficiency resistance	Single fixed R at which the turbine reaches greatest efficiency across all tested flow rates
Q_{eff}	Peak-efficiency flow rate	Flow rate at which that occurs, the turbine's natural sweet spot
R_{max}	Asymptotic optimal resistance	Used when the minimum tested flow is also the most efficient (no interior sweet spot). The asymptote that optimal R approaches as Q increases

Procedure

1. **Measure** R_{coil} with a multimeter. Optimal load will be higher than R_{coil} because of inductive reactance (ωL). True optimum is found in step 3.

2. **Open-circuit baseline.** Sweep flow from zero with leads open; record Q_{start} and (optionally) ΔP_{oc} vs. Q.
3. **Efficiency sweep.** At each Q, sweep R_{load} and compute $\eta = (V_{load}^2/R_{load})/(\Delta P \cdot Q)$. One curve per flow rate.
4. **Identify the peak.**
   - **Interior sweet spot found:** that (R, Q) is (R_{eff}, Q_{eff}). Use R_{eff} as the fixed load.
   - **Min flow rate is most efficient:** plot optimal R vs. Q; it asymptotes to R_{max}. Use R_{max} as the fixed load. Operating point is then driven by pressure budget or power need.

Sanity check: the power-vs-flow curve should begin to flatten near Q_{eff}, the same electromagnetic saturation that makes high-flow operation wasteful.

Flatness observation: R within a reasonable range around the optimum produces only modest changes in power and efficiency. A well-chosen fixed load captures nearly all the available benefit (verified per-turbine in §5).

Fixed Load Selection Rule

Turbine Type	Fixed Load	Notes
Has interior Q_{eff}	R_{eff}	Best efficiency at the natural sweet spot
Min flow = Q_{eff}	R_{max}	Selected by pressure budget or power need

Pre-Programmed Dynamic Load (Optional)

A lookup table indexed by flow rate can switch between a small set of resistors, populated from bench data (not real-time tracking). Adds ~$ 2 to $ 5 BOM. Worth considering when the operating range has a large span or if the optimal R shifts dramatically across its range. For most 0 to 4 GPM applications, a fixed R_{eff} or R_{max} captures the large majority of the available benefit, as shown in §5.1 Step 2.

4.3 Standard Output: Cost / Reward Chart

Once the fixed load is chosen, each turbine is summarized by a **normalized Cost / Reward chart**:

- **Cost:** ΔP vs. Q (hydraulic cost imposed on the system)
- **Reward:** Power vs. Q at the chosen fixed load (electrical output)

Both are expressed as percent of their turbine-specific maximum, so they share one axis. Conversion factors back to engineering units are given per-turbine in the caption.

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5. Test Results

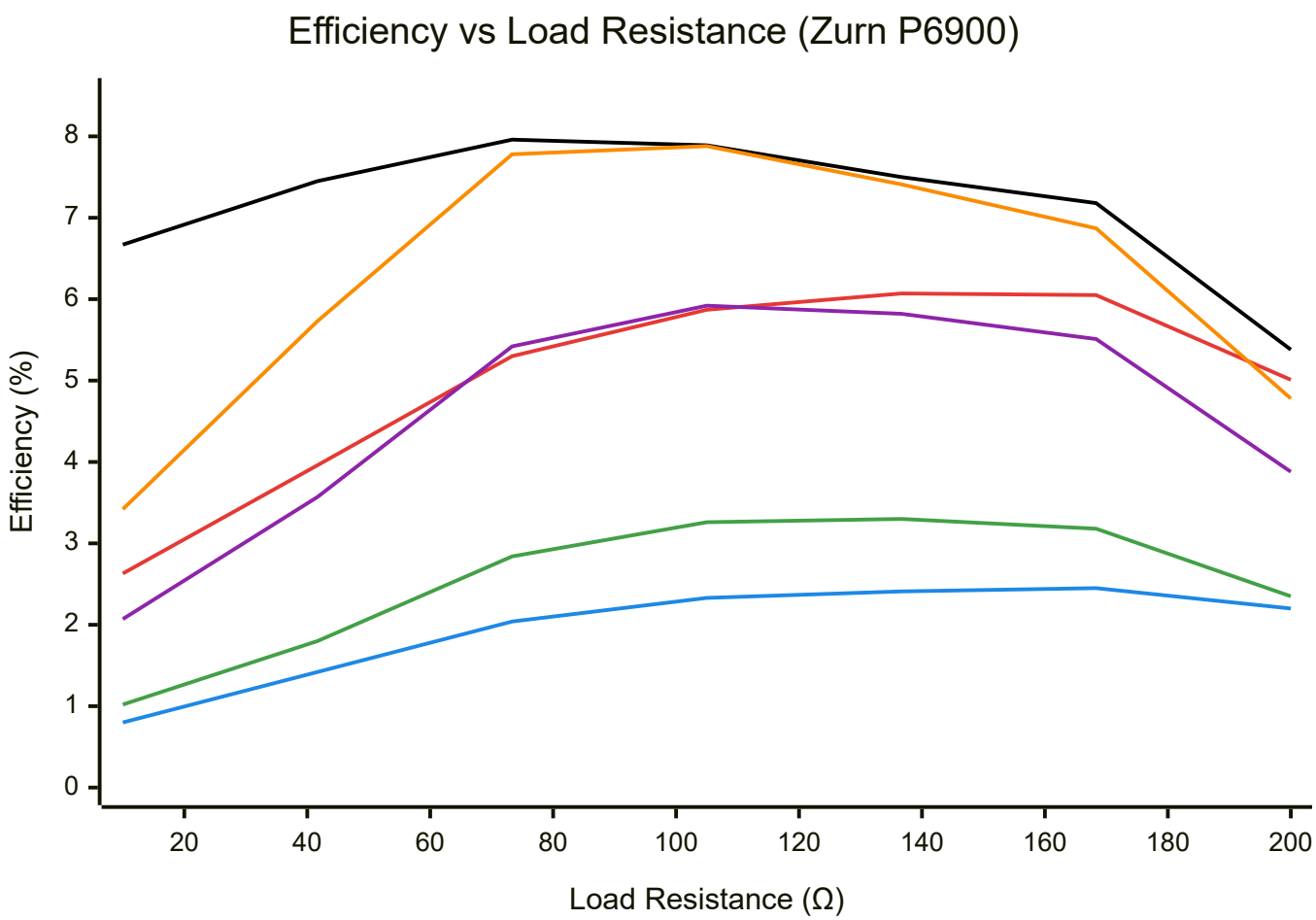
Two units demonstrate both branches of the selection rule.

5.1 Zurn P6900, Interior Sweet Spot (R_{eff} Case)

Primary demonstration unit with complete bench data (full Q-sweep × full R-sweep). Natively regulated DC; results below were taken with battery and regulator disconnected, tapping the raw coil leads. Concept demonstrator (clean ΔP curvature, clear interior Q_{eff}), not a product recommendation.

Measured: R_{coil} = 3.6 Ω.

Step 1: Efficiency vs. Load Resistance



Legend:

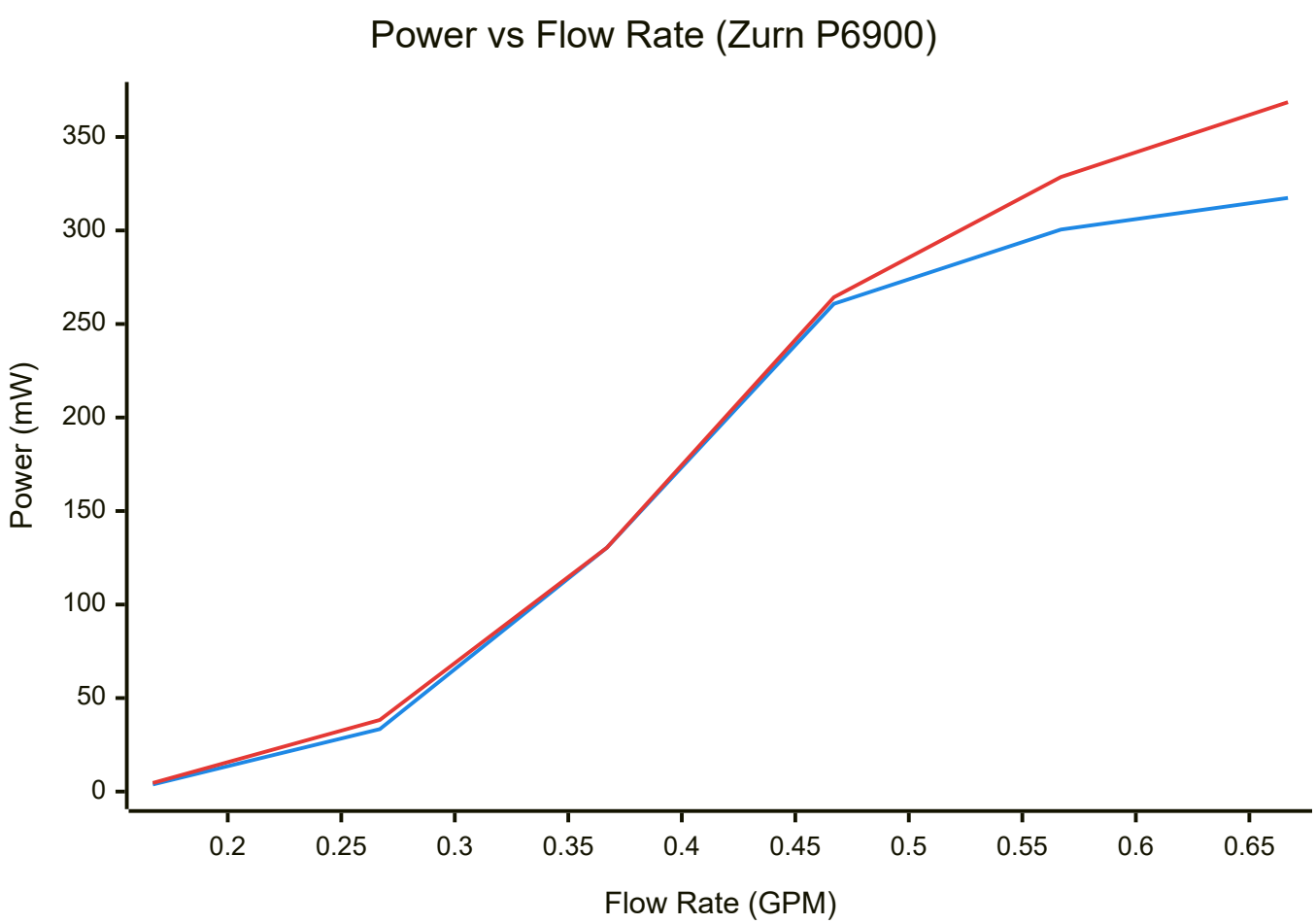
	Flow Rate	Peak η
	0.17 GPM	2.5%
	0.25 GPM	6.1%

	Flow Rate	Peak η
■	**0.33 GPM (Q_{eff})**	**8.0%**
■	0.42 GPM	7.9%
■	0.50 GPM	5.9%
■	0.67 GPM	3.3%

The global maximum is on the Q = 0.33 GPM curve at R = 80 Ω, $\eta \approx$ 8.0%. **R_{eff} = 80 Ω, Q_{eff} = 0.33 GPM.** Notice the flatness around 80 Ω: small R changes produce only marginal returns.

Step 2: Static R_{eff} vs. Dynamic Optimal R

With R_{eff} fixed, power vs. flow can be compared against the theoretical ceiling of a continuously-optimized load. The two lines touch at Q_{eff} and separate as Q grows because the optimal R shifts from 80 Ω toward 100 Ω at higher flow. **The flow range over which the two curves track each other closely is the turbine's effective working range under a fixed load**; outside that band, dynamic load-matching would start to provide meaningful gains.

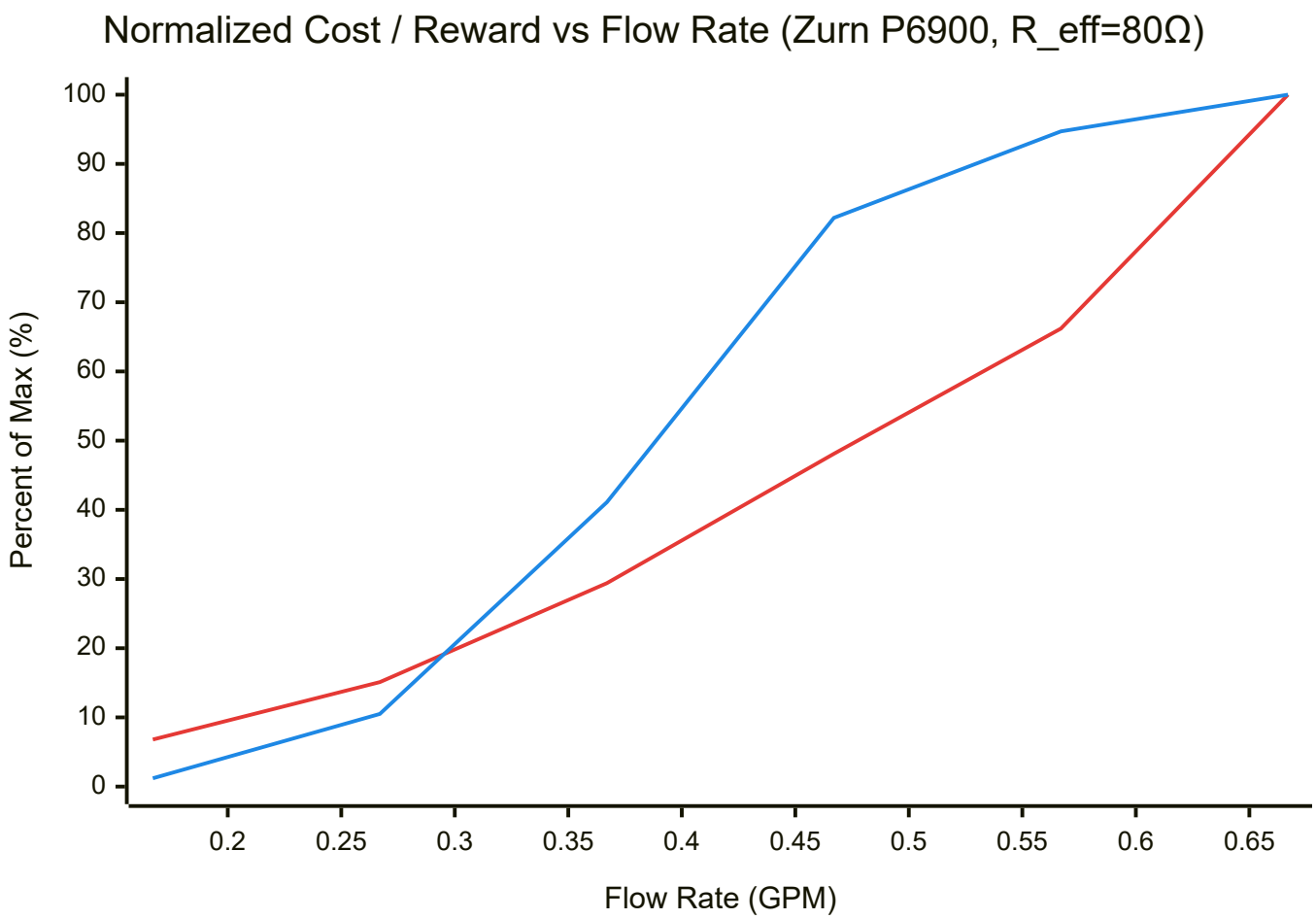


Legend:

	Load Strategy
🟦	Static load at R_{eff} = 80 Ω
🟥	Dynamic optimal R per Q (theoretical max)

The static-load curve captures the large majority of the theoretical maximum across the full operating range, with the largest gap (~14%) at Q = 0.67 GPM, but would continue growing at higher flow rates (limited flow capability on test bench). **For a fixed-load design R_{eff} is robust**; dynamic load chasing isn't worth the BOM and complexity for this margin on the Zurn.

Step 3: Cost / Reward Summary



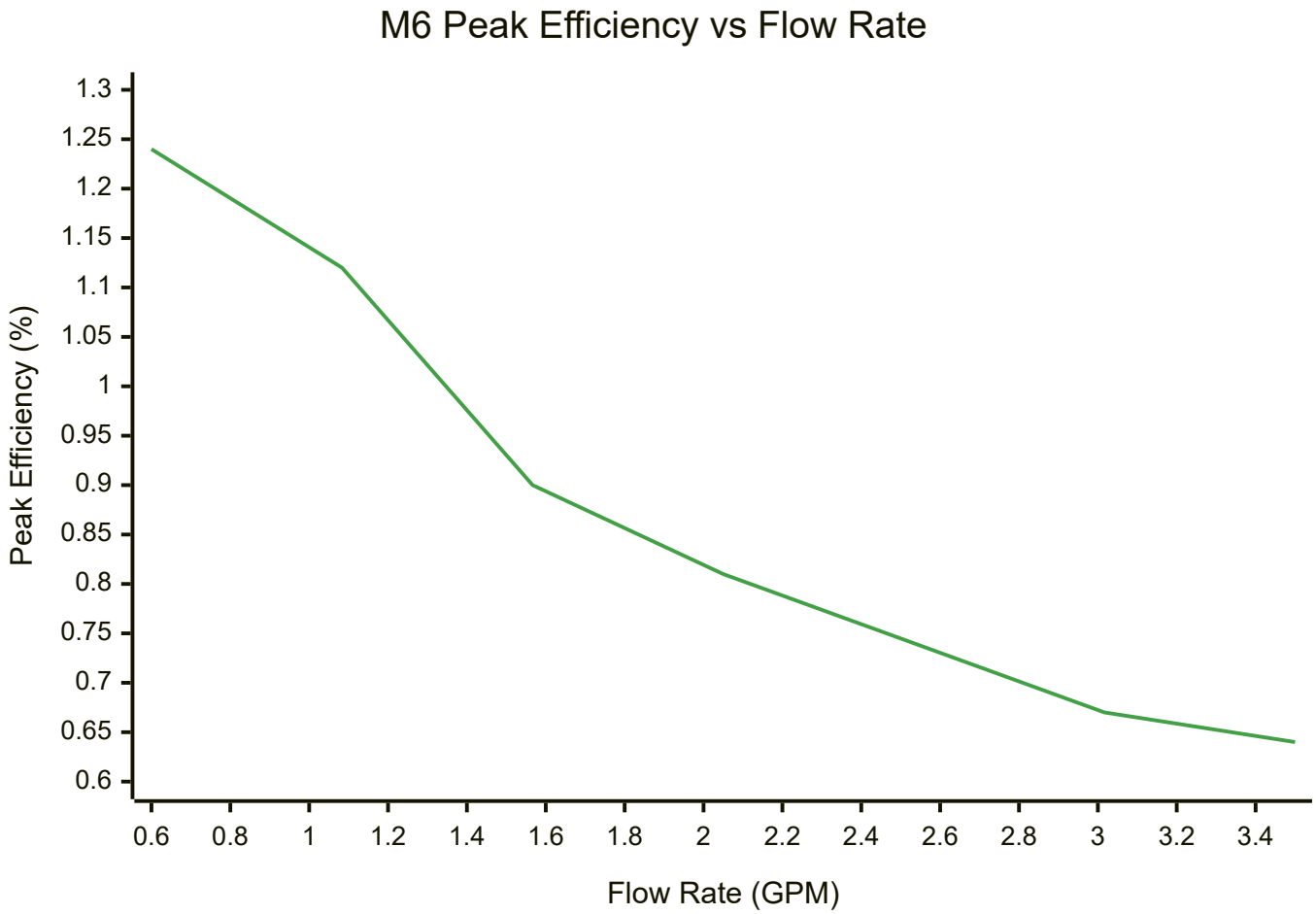
Legend:

	Series	Conversion
🟥	Cost (ΔP as % of 38.5 PSI max)	ΔP [PSI] = cost% × 0.385
🟦	Reward (Power at R_{eff} as % of 317.4 mW max)	P [mW] = reward% × 3.174

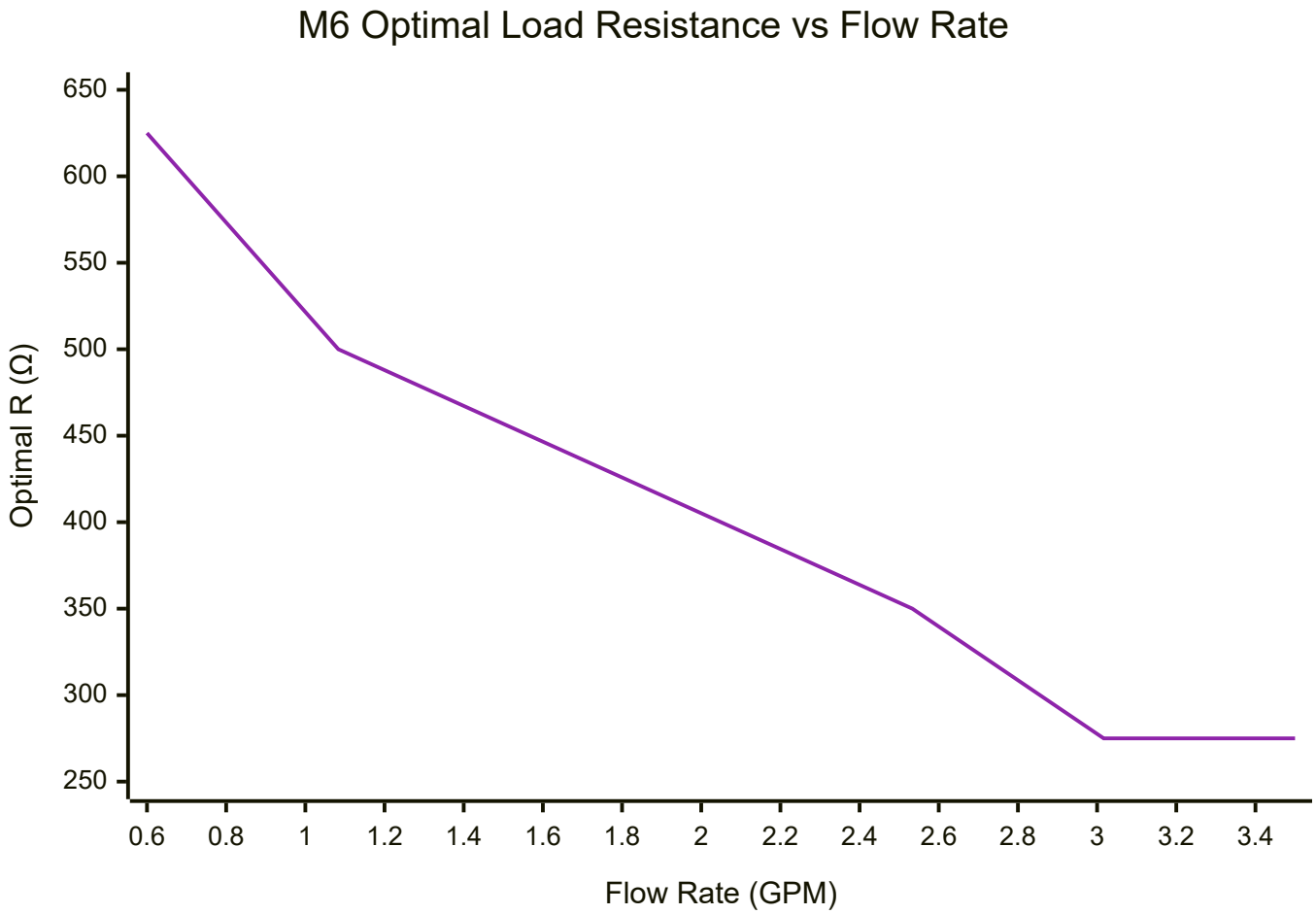
Reward rises steeply through Q_{eff} = 0.33 GPM, hitting 41% of max while Cost is still at 29%. Above Q_{eff}, Reward flattens while Cost keeps climbing; the crossover near Q = 0.42 GPM (Cost 48%, Reward 82%) marks where additional flow buys proportionally less electrical return. Lines are suspected to keep diverging at higher flow rates.

5.2 M6 Axial Propeller, Asymptotic Case (R_{max})

Full R-sweep at seven flow rates (0.60 to 3.50 GPM, R_{coil} = 161.6 Ω) shows **no interior efficiency peak**. Efficiency is highest at the lowest tested flow and decreases monotonically, characteristic of axial-propeller geometry.



Because no single flow rate is best, the load is chosen by looking at the optimal R at each Q. Optimal R decreases with increasing Q as shown below.



$R_{max} \approx$ **275 Ω**, where R begins leveling off (or asymptoting, further testing needed) is the fixed load selected for the M6. Final operating-point selection is driven by pressure budget and power need rather than a sweet-spot flow rate. M6 peak power ranges from ~3 mW at 0.60 GPM (ΔP ≈ 1 PSI) to ~490 mW at 3.50 GPM (ΔP ≈ 50 PSI), with efficiency steadily degrading.

5.3 Side by Side

	Zurn P6900	**M6 Axial Propeller**
Interior Q_{eff}?	Yes (0.33 GPM)	No
Fixed load	R_{eff} = 80 Ω	$R_{max} \approx$ 275 Ω
Operating point chosen by	Q_{eff}	Pressure budget / power need
Static-vs-dynamic gap	Negligible inside operating range	n/a (no interior peak)

These two demonstrate both branches of the selection rule. All other turbines in the catalogue follow the same workflow as bench data becomes available.

6. Sensing from a Generator

A turbine generator can double as a flow sensor because output is proportional to rotor speed, which is proportional to flow rate (and voltage in certain cases, power less reliably). This section covers how to extract Q from the generator and which output types support which method.

6.1 Generator Output Types

Output Type	Built-in Circuit	Sensing from electrical terminals?
Raw AC	None	Yes, best option (frequency or voltage-at-load)
Unregulated DC	Diode bridge	Yes, voltage-at-load; ripple frequency also usable
Regulated DC	Voltage regulator	**No**, flow info erased at output. Sensing requires opening the housing for an internal Hall (custom assembly only)

Rectification

Generator Type	Rectification
Single-coil, 2-wire	Bridge rectifier + smoothing cap
3-phase hub generator	3-phase full-wave (6 diodes)
Dual-coil, 4-wire	Separate rectification per coil, then combine

6.2 Sensing Methods, in Priority Order

1. Hall sensor inside the turbine body

Reads the rotor magnet directly, same as a dedicated flow meter (§2). Clean digital pulse train, frequency ∝ RPM, load-independent. With an A1171 (~15 to 42 μW) it's the most power-efficient method and cheaper than any PCB chain (op-amp + comparator + filter + ADC) trying to recover Q from the raw generator output. Preferred where physically possible, and the only option for regulated-DC units (after opening the housing).

2. AC or DC ripple frequency

Raw AC zero-crossings or unregulated DC ripple are proportional to Q and load-independent, same as the Hall pulse train. Avoid heavy downstream filtering; it kills ripple amplitude. Useful when the turbine can't fit an internal Hall sensor.

3. Voltage at a fixed load

Voltage tracks RPM and is insensitive to inlet pressure (unlike power). Works on raw AC or unregulated DC. Linear at low to moderate Q; saturates at high Q and needs a per-turbine lookup table past that point. Better than power-at-load, worse than frequency.

4. Power at a fixed load and pressure

$P \propto Q^2$ only at constant ΔP. Real plumbing pressure fluctuates with building demand, so the same Q can read differently. Cheapest option (load resistor + voltage measurement) and fine where inlet pressure is stable or accuracy is modest. Not recommended as the primary method when frequency-based options are available.

6.3 Open-Circuit Sensing Mode

With no load ($R_{load} = \infty$), electromagnetic braking is removed and the turbine spins more freely, pushing minimum detectable flow below the loaded Q_{start}. Flow is read from AC frequency or a Hall sensor. Requires a MOSFET to disconnect the load and a stored energy source (battery or supercap) to power the sensing circuit during this mode.

The system can in principle switch dynamically (open-circuit at very low flow for sensitivity, closed-circuit at higher flow for power generation), but the value is unclear in practice: typical efficiencies are under 10% (often under 5%), and disconnecting the load only reduces ΔP by an estimated 5 to 10%. Whether the gain in low-flow sensitivity justifies the added MOSFET, control logic, and stored-energy buffer is **a question for bench validation per turbine**, not a default architecture.

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7. System Architectures

7.1 Output Configurations

Every rotary flowsense decision resolves to one of the following. "Handles 0-flow events" means the system stays responsive when no water is flowing; "Flow minimum with turbine" is the lowest Q at which the turbine path can sense and/or power the system.

#	Configuration	Sensing	Power	Handles 0-Flow Events	Flow Minimum
1	Flow meter + plug in	Dedicated meter (GEMS + A1220)	Wall Socket	No	Meter sets the floor, ≈ 0.08 GPM
2	Flow meter + battery	Dedicated meter (GEMS + A1171)	Battery	Yes (sleep)	Meter sets the floor, ≈ 0.08

#	Configuration	Sensing	Power	Handles 0-Flow Events	Flow Minimum
					GPM
3	Generator as sensor	Power at fixed load, or AC / ripple frequency	Turbine output	No	Loaded Q_{start} (or minimum Q to power circuit)
4	Generator + internal Hall + cap or battery	Internal Hall (load-independent)	Turbine + small battery	Yes (battery)	Open-circuit Q_{start} (lower than loaded)
5	Paired flow meter + generator + battery	Dedicated meter	Turbine + battery trickle-charge	Yes (battery)	Meter floor (≈ 0.08 GPM)

7.2 Operating States & Sleep

The selection of a flowmeter depends heavily not just on the water pressure and flow rate, but the frequency and duration of use. A sink may be used for a few seconds at a time or be unused for an entire day. Measuring a flow rate of 0 takes the same amount of effort measuring any other flow rate, and a turbine can only provide power while its being used. Thus, the active / sleep distinction often dominates total energy budget.

State	Condition	Power Draw	Power Source
Active, generating	Water flowing, load connected	MCU + Hall + radio (5 to 50 mW typical)	Turbine; charges battery in parallel
Active, sensing only	Water flowing, load open	MCU + Hall only (~0.1 to 1 mW)	Battery; braking minimized for low-flow sensitivity
Sleep	No flow	MCU deep sleep (~1 to 100 μW)	Battery only

Switching between Generating and Sensing-Only requires a MOSFET load-disconnect and a battery; it is the same architectural question raised in §6.3 and should be validated per turbine before committing to it.

Wake from sleep:

- **Flow-triggered wake (preferred):** low-power comparator monitors Hall signal or raw AC at ~1 to 10 μA quiescent. A detected pulse wakes the MCU.

- **Periodic polling:** MCU wakes on a timer and checks for flow. Simpler firmware but wastes energy during long no-flow periods.

> Specific circuit topology for sleep switching and battery management is out of scope. Design principle: size the battery to cover expected sleep duration at sleep current; size the turbine to replenish it within a normal active flow event.

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8. Turbine Selection Framework

8.1 Input Criteria

The selection process matches a product's **water-usage profile** and **power-needs profile** against each turbine's measured Cost / Reward function.

Input	Description	Default / Notes
Flow range (GPM)	Expected operating range; assume normal distribution	Required
Flow frequency	How often the fixture sees flow (e.g., bursts/day)	Required
Flow duration	Typical event length (seconds to minutes)	Required
Power tier	Sensor only, low-power display, BLE / radio connectivity	Required
Sensing?	Yes / No (if yes what are accuracy needs)	Required
Pressure budget (PSI)	Maximum allowable ΔP	Default 50 PSI. Legal min 25, legal max 80
Price budget	Max unit cost	Optional; if unknown, use as a visualization axis

Power tier menu (rough numbers, refine per product):

Tier	Typical Continuous Power	Notes
Flow sensor only	~0.1 to 1 mW	MCU + Hall only
Sensor and other sink functions	~5 to 10mW	Solenoid and/or hand sensor

Tier	Typical Continuous Power	Notes
Low-power display	~10 to 20 mW	Adds LCD or e-ink
BLE / radio connectivity	~20 to 50 mW peak (bursty)	Needs storage to buffer transmits

8.2 Selection Logic

Each turbine has a **Cost function** (ΔP vs. Q) and a **Reward function** (power at fixed load vs. Q), measured on the bench. Selection is a matching exercise:

1. **Filter by flow range and pressure budget.** Use the Cost portion of each turbine's Cost / Reward chart. Eliminate turbines whose ΔP at the expected flow exceeds budget, or whose usable range doesn't overlap the product's flow distribution.
2. **Match Reward to power tier.** For the expected flow, frequency, and duration, integrate Reward against the power tier requirement. A short, frequent flow event into a BLE-connected node implies large transient draw and benefits from a turbine that hits high power quickly; a long, infrequent event into a sensor-only PID can tolerate a turbine that takes longer to reach steady-state.
3. **Apply sensing requirement.** If sensing is required, confirm the turbine can physically fit a Hall sensor (or that frequency-based sensing on the raw output is acceptable). Regulated-output turbines require Hall + custom housing; power-based sensing alone is not recommended, especially with accuracy constraints (see §6.2).
4. **Apply price filter.**

Step 2 above (matching the turbine's Cost / Reward function to a specific product's water-usage and power-needs profile) is the **next step** in this work, executed once the full catalogue has been bench-characterized. The framework is set up; the data fills in turbine-by-turbine.

8.3 Visualization

Plot all turbines on a 2D scatter with axes chosen from the unknowns or trade-offs of interest (e.g., peak power vs. ΔP at expected Q, or price vs. peak power). Each point is one turbine. This is the visual companion to the matching process in §8.2 and the most useful single artifact when comparing options for a new product.

A list of suitable applications is built up under each turbine in the catalogue as candidates are eliminated or confirmed by the matching process.

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9. Application Examples

Two Watts product lines are used as worked examples: **Bradley** commercial restroom faucets and **Haws** bottle-filling stations. Both have flow-sensing and/or IoT needs that justify going through the selection process, and both are good candidates for a future-proofed self-powered architecture.

9.1 Bradley, Commercial Restroom Faucet

Context: Bradley restroom faucets currently lack flow sensing and IoT capability. Adding usage monitoring, anomaly detection, and volume tracking without wired power or battery replacement is the goal. Restroom flow events are short (seconds), frequent (many per day), and clustered around peak hours.

Input	Value
Flow range	0.5 to 1.5 GPM
Flow frequency	TBD
Flow duration	Short (~5 to 15 s per event)
Power tier	Up to BLE connectivity
Sensing?	Yes
Pressure budget	50 PSI (default)
Price budget	TBD

Configuration is open: a turbine with internal Hall sensing and a small buffer battery (config 4 in §7.1) is the most natural fit for IoT, but a "power + frequency sensing" path on a turbine with raw AC output is also viable if the candidate turbine's frequency signal is clean enough at expected flows. Specific recommendation deferred until catalogue R_{eff} / Q_{eff} data is complete.

9.2 Haws, Bottle-Filling Station PID and IoT Upgrade

Context: Haws bottle-filling stations use replaceable filters with rated gallon limits. A **Performance Indicating Device (PID)** is required under NSF/ANSI 42 (§5.3.2) and NSF/ANSI 53 (§6.3.2). PID requirements:

- Track **actual water volume** by physical displacement (rotating turbine or positive-displacement). Time-based estimation does not qualify.
- Built-in, not aftermarket.
- Operates automatically; manually resettable after filter replacement.
- Non-adjustable end-of-life setpoint.

- Distinct visual or audible end-of-life indicator (NSF doesn't mandate a specific color but requires a distinct warning state).

Per NSF/ANSI 53, a certified PID allows filter life testing to 120% of rated capacity rather than 200% without one.

Current solution: **Digiflow 8100T-22** ($17.33, 2x AA, ~3-year life replaced with the filter 2,000 gallons, LCD display).

Two goals at different price points:

Goal	Description
Cost reduction	Compliant rotary-turbine PID for less than $17.33
IoT upgrade	Add cloud connectivity using flow-generated power, beyond what 2x AA supports

Input	Value
Flow range	0.5 to 1.6 GPM
Flow frequency	TBD
Flow duration	Medium (15-30 seconds)
Power tier	PID only (sensor + LCD): ~3 mW active / ~45 μW sleep. IoT tier: BLE, TBD
Sensing?	Yes (cumulative gallon count required)
Pressure budget	50 PSI (default; confirm Haws inlet spec)
Price budget	< $17.33 cost-reduction; relaxed for IoT

Flow range overlaps Bradley closely so the same candidate turbines apply. The cost-reduction goal is sensing-dominant and could be served by config 1 or a low-cost turbine in config 3. The IoT goal requires sustained BLE plus trickle-charging during sleep, pointing toward config 4 or 5. As with Bradley, final selection comes out of the matching exercise in §8.2.

Appendix A: Catalogue: Flow Meters

Meter	GEMS 238600 + A1220	GEMS 238600 + A1171	Gredia FS200A	Sea YF-S201	Uxcell	Sika VY10	Auk Mueller MT5	Digiflow 8100T-22
Picture	**take picture**	**build**						
Category	Custom Build	Custom Build	Off-the-shelf	Off-the-shelf	Off-the-shelf	Off-the-shelf	Off-the-shelf	Off-the-shelf, Display Included
Manufacturer	GEMS Setra (Plainville, CT)	GEMS Setra (Plainville, CT)	Gredia (China)	Sea (China)	Uxcell (China)	Sika (Germany)	A.u.K. Muller (Germany)	Savant / Digiflow
Turbine Type	Axial	Axial	Radial	Radial	Radial	Axial	Axial	Radial
Admission	Fully enclosed	Fully enclosed	Partial	Partial	Partial	Full	Full	Partial
Min Flow (GPM)	0.08	0.08	~0.13	~0.26	~0.1-0.15	0.16-0.18	0.05	0.21
Max Flow (GPM)	2.0 (tested to 4.0)	2.0 (tested to 4.0)	7.9	7.9	7.9	7.9	3.96	3.96
Signal Sensor	Allegro A1220	Allegro A1171	Built-in Hall	Built-in Hall	Built-in Hall	Built-in Hall (NPN open collector)	External detector clip	Built-in Hall
Output	Hall pulse (freq)	Hall pulse (freq)	Hall pulse	Hall pulse	Hall pulse	Square wave freq	Hall pulse (open drain)	LCD display
Active Power	~12 mW @ 3V	~42 µW	~75 mW @ 5V	~75 mW @ 5V	~75 mW @ 5V	45-240 mW	N/A	~3 mW
Sleep Power	N/A	~15 µW	N/A	N/A	N/A	N/A	N/A	~45 µW
Accuracy	2%	2%	±1–2%	±10%	N/A	±1% of range	±2% of range	±5%
NSF / WRAS	NSF 61, WRAS	NSF 61, WRAS	No	No	No	NSF 61, 372, WRAS, KTW, ACS	NSF approved	NSF 61/9
Max Temp	160°F	160°F	248°F	N/A	N/A	194°F (brass) / 158°F (plastic)	194°F media	170°F
Max Pressure	150 PSI	150 PSI	174 PSI	254 PSI	N/A	232 PSI (brass) / 145 PSI (plastic)	232 PSI	112 PSI
Unit Cost	~$12	~$12	$7.99	$3.92	$4.29	$9.32 (5K vol)	$16.50-$17.60	$17.33
Materials	ABS cage, Nylon 6 turbine, PEEK bearings	ABS cage, Nylon 6 turbine, PEEK bearings	ABS	Nylon	Plastic	Brass CW617N or Noryl 30% GF; SS shaft; hard ferrite magnet	PPSU body, SS shaft, ruby bearings, EPDM seals	PA66+50% GF, 304 SS axle, EPDM, ABS display
Fitting	Custom 3/8" FNPT	Custom 3/8" FNPT	G1/2"	G1/2"	G1/2"	G1/2", G3/4", 1/2" NPT, QuickFasten	Cable (3x0.25mm^2 PVC)	3/8" BSPP male
Notes	Proven in Moen U & Delta Essa. Best low-flow sensitivity.	~800x less power. Best for battery/harvesting.	Tested in-house.	Cheapest. Poor accuracy. Tested in-house.	Currently unavailable. Tested in-house.	Sapphire/PEEK bearings. Integrated flow straightener. 5kΩ pull-up needed. Tested in-house.	Two-part: turbine + detector clip.	Horizontal only. 2xAA. Sleep 15 µA.

Appendix B: Catalogue: Generators

Meter	Sea DB-3682	Sea DB-168 AC20V	Sea DB-168 DC5V	M6 Connector 5V	Zurn P6900-GEN	Sloan EFP40A	Toto EcoPower Insert (THP3275)	Toto TH559EDV462 (0.5 GPM Dynamo)	Toto TH559EDV561 (1.0 GPM Dynamo)
Picture							take picture		
Category	Generator	Generator	Generator	Generator	Generator (+ battery)	Full Assembly	Generator	Generator	Generator
Manufacturer	Sea / Dijiang (Foshan, China)	Sea / Dijiang (Foshan, China)	Sea / Dijiang (Foshan, China)	Generic (China)	Zurn	Sloan	Toto	Toto	Toto
Output Type	Raw AC	Raw AC	Regulated DC	Raw AC (tested via raw coil tap)	Regulated DC (Raw AC tested via raw coil tap)	Regulated DC	Raw AC	Raw AC	Raw AC
Turbine Type	Radial	Radial	Radial	Axial	Radial	Radial	Radial	Radial	Radial
Admission	Full	Partial	Partial	Full	Full	Full	Full	Full	Full
Flux Type	Axial	Radial	Radial	Radial	Axial	Axial	Axial	Axial	Axial
Min Flow (GPM)	~0.53	~0.53	~0.53	~0.6 (tested)	~0.17 (tested; 0.5 spec)	~0.5	N/A (not tested)	~0.35-0.5	~0.35-0.5
Max Flow (GPM)	~3.96	~2.64	~2.64	~3.5 (tested)	~1.5	~2.2	N/A (not tested)	~0.5	~1.0
Output	Raw AC 0-20V	Raw AC 0-20V	5V DC	5V DC	Internal battery + storage (untested)	Internal + battery backup (untested)	AC, 2-wire	AC, 9-35V	AC, 4-wire (dual coil)
Power Production	Up to 3W	Up to ~1W	Up to 0.75W	~490 mW @ Q = 3.5 GPM (tested)	~369 mW @ Q = 0.67 GPM (tested)	~mW range	~0.5W (est.)	~0.5W	~0.5W
Coil Type	Suspected hub generator, teardown required	Hub generator	Hub generator	Gyro (ring-in-ring)	Claw pole	Claw pole	Belt (wrapped)	Belt (wrapped)	Belt (wrapped)
Coil Config	TBD	Multi (3-phase)	Multi (3-phase)	—	Single	Single	Single (2-wire)	Single	Dual
R_eff (Ω)	TBD	TBD	TBD	N/A (R_max = 275)	**40**	TBD	TBD	TBD	TBD
Q_eff (GPM)	TBD	TBD	TBD	N/A (min flow = Q_eff)	**0.33**	TBD	TBD	TBD	TBD
Min Starting Pressure	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Fitting	G1/2"	G1/2"	G1/2"	G1/2" + M6	1/2"-14 NPSM	Proprietary (undersink box)	Non-separable (in faucet spout)	Proprietary (undersink)	Proprietary (undersink)
Max Temp	140°F	176°F	176°F	N/A	N/A	N/A	N/A	N/A	N/A
Max Pressure	254 PSI	174 PSI	174 PSI	N/A	80 PSI (rec.)	125 PSI	N/A	108 PSI	108 PSI
NSF	No	No	No	No	Yes	Yes	Yes (ASME, CSA)	Yes (ASME, CSA)	Yes (ASME, CSA)
Unit Cost	$13	$3.50 ($4 w/ NTC)	$3.50 ($4 w/ NTC)	$8	~$50-100+	Not sold seperately	Not sold separately	~$75+	~$75+
Tested?	No	No	No	**Yes (retest)**	**Yes (retest)**	Partial	Partial	Partial	No